

Chapter 3

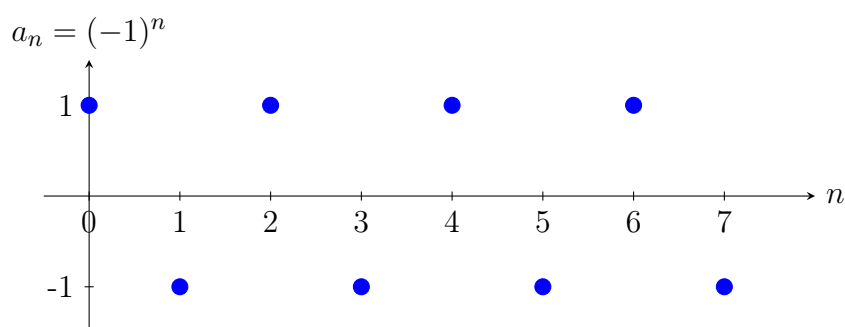
Sequences

Definition

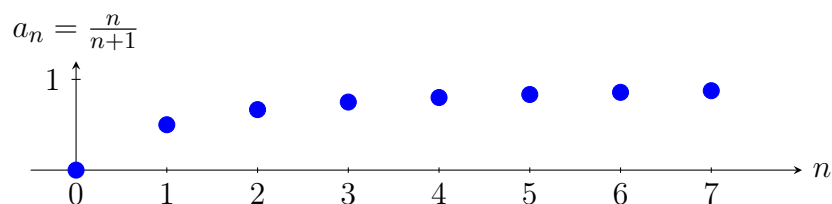
A **sequence** of real numbers $a_0, a_1, a_2, \dots$ is an assignment of a real number a_n to each natural number n (or to each n in some subset of $\mathbb{N}$). We denote it

$$\{a_n\}_{n \in \mathbb{N}} \quad \text{or} \quad \{a_n\}_{n=0}^{\infty}$$

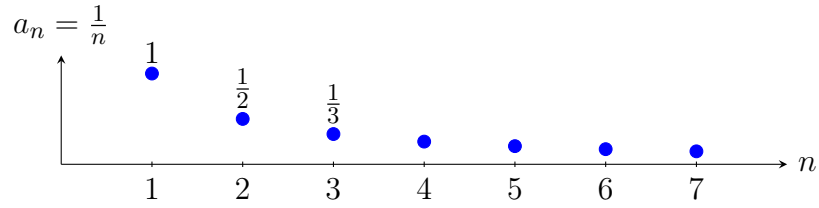
Example 3.1: 1. We start with the graph of the sequence $\{a_n\}_{n \in \mathbb{N}}$ given by the rule $a_n = (-1)^n, n \in \mathbb{N}$. This sequence is simply given by $(-1)^n = \begin{cases} 1 & \text{when } n \text{ is even} \\ -1 & \text{when } n \text{ is odd} \end{cases}$ and looks as follows:



2. Here is the graph of $a_n = \frac{n}{n+1}, n \in \mathbb{N}$:



3. Here is the graph of $a_n = \frac{1}{n}$, defined only for $n \in \mathbb{N}_+$:



4. Not every sequence is given by an explicit formula in n ; some are defined **recursively**, in terms of earlier terms. A classical example is the **Fibonacci sequence**, defined by $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

FIBONACCI (LEONARDO OF PISA, C. 1170–1250)

Fibonacci was born in Pisa around 1170. As a boy he travelled with his merchant father to Bugia, in present-day Algeria, where he was educated and learned the Hindu-Arabic numeral system from his teachers there. Recognizing how much easier this system made calculation compared to Roman numerals, he wrote the Liber Abaci (1202) to bring it to Europe – an exchange of knowledge across the Mediterranean that reshaped European arithmetic and commerce. The sequence above appears in that book as an example problem, and was only named after him centuries later.

Let $N \in \mathbb{N}$. Then:

- A sequence $\{a_n\}_{n \in \mathbb{N}}$ is said to be **(monotonically) increasing on $\{N, N+1, \dots\}$** if

$$\forall n \geq N, \quad a_n \leq a_{n+1}.$$

- A sequence $\{a_n\}_{n \in \mathbb{N}}$ is said to be **strictly increasing on $\{N, N+1, \dots\}$** if

$$\forall n \geq N, \quad a_n < a_{n+1}.$$

- A sequence $\{a_n\}_{n \in \mathbb{N}}$ is said to be **(monotonically) decreasing on $\{N, N+1, \dots\}$** if

$$\forall n \geq N, \quad a_n \geq a_{n+1}.$$

- A sequence $\{a_n\}_{n \in \mathbb{N}}$ is said to be **strictly decreasing on $\{N, N+1, \dots\}$** if

$$\forall n \geq N, \quad a_n > a_{n+1}.$$

Example 3.2: 1. The sequence $a_n = \frac{1}{n}$, $n \in \mathbb{N}_+$, is strictly decreasing.

2. The sequence $a_n = \frac{n}{n+1}$, $n \in \mathbb{N}$ is strictly increasing.

3. The sequence $a_n = (-1)^n$, $n \in \mathbb{N}$ is neither increasing nor decreasing.

3.1 Subsequences

Considering the sequence $a_n = (-1)^n$, $n \in \mathbb{N}$,

$$1, -1, 1, -1, 1, -1, \dots, (-1)^n, (-1)^{n+1}, \dots,$$

we see that if we look only at even indices the resulting sequence always attains the value 1, whereas if we only look at odd indices the resulting sequence always attains the value -1 . Those two new sequences are examples of *subsequences* of the original sequence. We will later see that it is often useful to restrict our attention to only some elements of a sequence, hence this notion is important.

Definition

A **subsequence** of a sequence $\{a_n\}_{n \in \mathbb{N}}$ is a new sequence $\{a_{n_k}\}_{k \in \mathbb{N}}$, where $\{n_k\}_{k \in \mathbb{N}}$ is an increasing sequence of natural numbers.

Thus, the subsequence of even indices is given by the choice

$$n_0 = 0, \ n_1 = 2, \ n_2 = 4, \dots, \ n_k = 2k, \dots$$

whereas the subsequence of odd indices is given by the choice

$$n_0 = 1, \ n_1 = 2, \ n_2 = 5, \dots, \ n_k = 2k + 1, \dots$$

Chapter 4

Functions

4.1 Definitions and examples

Definition

Let X and Y be two sets. A **function f from X to Y** is a rule that associates to any element $x \in X$ at most one element $y \in Y$. The subset of elements in X to which f associates an element in Y is called the **domain of f** and is denoted $\text{dom}(f)$. We write

$$f : \text{dom}(f) \subseteq X \rightarrow Y.$$

For $x \in \text{dom}(f)$, the element $y \in Y$ associated to it by f is called the **image of x under f** and is denoted $y = f(x)$. We often write

$$f : x \mapsto f(x).$$

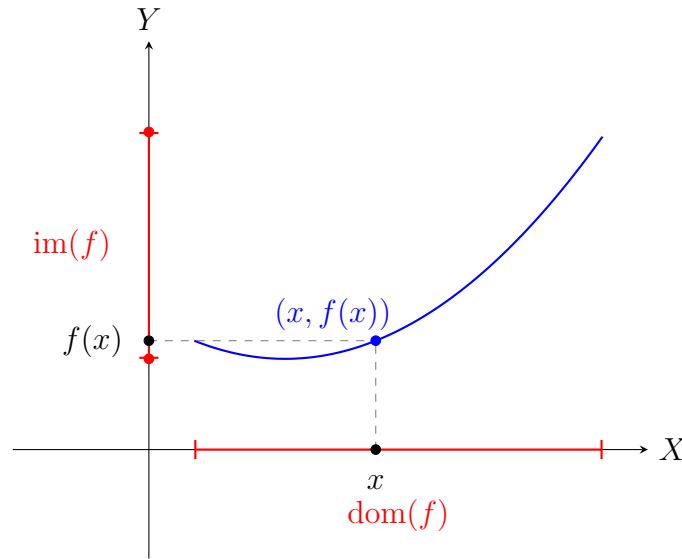
The subset of Y of all images of elements in X is called the **range of f** and is denoted:

$$\text{im}(f) = \{y \in Y \mid \exists x \in \text{dom}(f), y = f(x)\}.$$

If $Y = \mathbb{R}$ we say that the function f is **real-valued**. Finally, the **graph of f** is the following subset of the Cartesian product $X \times Y$:

$$\Gamma(f) = \{(x, f(x)) \in X \times Y \mid x \in \text{dom}(f)\}.$$

Here is an example of how we can visualize these properties:



Example 4.1: Some notable examples for $f : \mathbb{R} \rightarrow \mathbb{R}$ include:

1. *Linear functions:* $f(x) = ax$ where $a \in \mathbb{R}$, $a \neq 0$. The graph is a straight line through the origin with slope a (the line cannot be vertical).
2. *Affine functions:* $f(x) = ax + b$ where $a, b \in \mathbb{R}$, $a \neq 0$. The graph is a straight line through the point $(0, b)$ with slope a (the line cannot be vertical).
3. *Quadratic functions:* $f(x) = ax^2 + bx + c$ where $a, b, c \in \mathbb{R}$, $a \neq 0$. The graph is a parabola.
4. *Square root:* $f(x) = \sqrt{x}$. This is the first function mentioned here whose domain is not $\mathbb{R}$: $\text{dom}(\sqrt{\cdot}) = \{x \in \mathbb{R} \mid x \geq 0\}$.
5. *Absolute value:*

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

6. *Sign function:*

$$f : \mathbb{R} \rightarrow \mathbb{Z}, \quad f(x) = \text{sign}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

7. *Ceiling* ('rounding up'):

$$f : \mathbb{R} \rightarrow \mathbb{Z}, \quad f(x) = \lceil x \rceil = \text{smallest } n \in \mathbb{Z} \text{ s.t. } n \geq x.$$

8. *Floor* ('rounding down'):

$$f : \mathbb{R} \rightarrow \mathbb{Z}, \quad f(x) = \lfloor x \rfloor = \text{greatest } n \in \mathbb{Z} \text{ s.t. } n \leq x.$$

9. *Sequences:* a sequence $\{a_n\}_{n \in \mathbb{N}}$ is a function

$$f : \mathbb{N} \rightarrow \mathbb{R}, \quad f(n) = a_n.$$